MILITARY SPECIFICATION

ACETAL, SHEETING
(FOR USE WITH AMMUNITION)

1. SCOPE

1.1 Scope.-This specification covers acetal resin in sheet form for use with ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification 21 - Ratings, Rules and Regulations, Association of American Railroads.

(Application for copies of these freight classification rules should be addressed to the Secretary of Operating Transportation Division, 59 Van Buren Street, Chicago, Illinois.)

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National Motor Freight Classification No. A-2, Naming Uniform Rating Rules and Regulations

(Copies may be obtained from the Issuing Officer, 1424 - 16th Street, N. W., Washington 6, D. C.).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 882 - Tests for Tensile Properties of Thin Plastic Sheeting
D 1004 - Tear Resistance of Plastic Film and Sheeting
D 1922 - Propagation Tear Resistance of Plastic Film and Thin Sheeting
D 792 - Specific Gravity and Density of Plastics by Displacement
D 646 - Thickness of Paper and Paperboard

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penna., 19103.)

3. REQUIREMENTS

3.1 Materials.-The acetal resin sheeting shall be virgin material with no reprocessed resin therein. The form shall be sheet.

3.2 Physical requirements.-The acetal resin shall comply with the requirements of Table I.

TABLE I

Properties	Requirements	Test Method
Specific Gravity, 23/23°C	1.410 ± .020	4.4.1
Elongation %, min.	10	4.4.2
Tensile Strength, (psi), min.	7,000	4.4.3
Tear initiation, lbs., min.	2.5	4.4.4
Tear propagation, gr., min.	2.0	4.4.5
Thickness, inches	.0015 ± .0003	4.4.6

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3.3 Workmanship.-The workmanship shall be first class in every respect. The acetal resin sheets shall be free of breaks, cracks, creases, holes, surface contamination, scratches, exudation, dirt, and bubbles. The edges shall be straight and no foreign material shall be present.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and Specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of rolls in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection provisions.

4.2.1 Lot formation.-The term "inspection lot" as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements.

4.3 Testing

4.3.1 Sampling.-A group of five (5) rolls shall be randomly selected from each lot. A sample shall be removed from each roll, tested and inspected for compliance with the requirements of this specification. No sample shall be taken from the first or last five yards of a roll. The samples shall be conditioned as specified in the ASTM Designation D618. The test results from each shall be averaged for reporting purposes. Failure of any individual sample to conform with the requirements of this specification shall cause rejection of the entire lot.

4.4 Test methods and procedures

4.4.1 Specific gravity, Major Defect, (see 6.3), Code No. 01001.-The specific gravity shall be determined in accordance with ASTM Designation D 792, Method A-1.

4.4.2 Elongation, Major Defect, Code No. 02001.-The elongation shall be determined in accordance with ASTM Designation D882, Method B.

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4.4.3 Tensile strength, Major Defect, Code No. 03001.-The tensile strength shall be determined in accordance with ASTM Designation D 882, Method B.

4.4.4 Tear initiation, Major Defect, Code No. 04001.-The tear initiation load shall be determined in accordance with ASTM Designation D 1004.

4.4.5 Tear propagation, Major defect, Code No. 05001.-The tear propagation load shall be determined in accordance with ASTM Designation D 1922.

4.4.6 Thickness, Major Defect, Code No. 06001.-The thickness shall be determined in accordance with ASTM Designation D 646.

5. PREPARATION FOR DELIVERY

5.1 Configuration.-The width of the sheeting shall be 36 $\pm$ 1/4 inches, supplied in 250 pound (min.) rolls on 3 inch cores.

5.2 Packing.-The sheeting shall be packaged in accordance with supplier's commercial practice. Protection shall be such as to prevent deterioration during shipment and subsequent intervals preliminary to use. The sheeting shall be packed for shipment in conformance with Consolidated Freight Classification Rules and container specifications for rail shipments, or with National Motor Freight Classification Rules and container specifications for truck shipments.

5.3 Marking.-Interior packages, and exterior containers, shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Intended use.-The sheeting is intended for use in laminated propelling charge bags for ammunition.

6.3 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.4 Celanese Plastics Company, 550 Broad Street, Newark, New Jersey, 07102, Film Designation XF-300 has been found satisfactory for meeting the requirements of this specification. Any other material proposed for this use should be submitted to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, Dover, New Jersey, 07801, for determination of compatibility with explosives.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 1395-A010

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